

Cost and Time Estimation

Sensor Data Collector (Mobile App)

Denis Schatzmann

June 11, 2026

1 Task

The Sensor Data Collector lets a user pick sensors from a select box, add them to a list (duplicates ignored), remove a marked sensor, and toggle data collection with a Start/Stop button. On stop the collected data is sent to a backend service. This report estimates size, effort, schedule and cost using the methods from the lecture: Function Point Analysis for size and COCOMO for effort and schedule.

2 Size: Function Point Analysis

Each function of the application is weighted with the low or average value from the IFPUG categories.

Function	Category	FP
Add sensor	External input	3
Remove marked sensor	External input	3
Start/Stop collection	External input	3
Send collected data to backend	External output	5
Populate sensor select box	External inquiry	4
List of selected sensors	Internal logical file	7
Total		25

Table 1: Function point count.

As a quick check, the widget point method counts four input widgets, three describing widgets and one list, so $8 \times 2 = 16$ function points. Both results place the application in the same small range, so the count of about 25 FP is plausible.

3 Effort: COCOMO

The function points are converted to lines of code with the language table. For a Java or Kotlin codebase the factor is 53 LOC per FP:

$$\text{LOC} = 25 \times 53 = 1325 \approx 1.33 \text{ KDL}.$$

Applying COCOMO in organic mode ($a = 3.2$, $b = 1.05$):

$$E_i = 3.2 \cdot 1.33^{1.05} \approx 4.3 \text{ person months.}$$

Because this is medical software, the required reliability cost factor is rated high (1.15), giving an effort adjustment factor $EF = 1.15$:

$$E = EF \cdot E_i = 1.15 \cdot 4.3 \approx 4.9 \text{ person months.}$$

4 Schedule and Staffing

The duration and the average number of staff follow from the organic formulas:

$$D = 2.5 \cdot E^{0.38} = 2.5 \cdot 4.9^{0.38} \approx 4.6 \text{ months,} \quad P = \frac{E}{D} \approx 1.1 \text{ persons.}$$

So one developer working about 4.6 months covers the work. With the effort partitioning for a small project the 4.9 person months split as:

Phase	Share	Person months
Design	19%	0.9
Implementation	63%	3.1
Integration and tests	18%	0.9

Table 2: Effort distribution for a small project.

5 Cost

Total cost combines personnel, software and hardware. At a fully loaded rate of about CHF 12'000 per person month, the personnel cost is

$$4.9 \times 12'000 \approx \text{CHF } 59'000.$$

Adding roughly CHF 4'000 for tooling licences, test devices and the cloud backend gives a total of about **CHF 63'000** for one developer over about **4.6 months**.

6 Accuracy

At the specification stage the estimate is only accurate within a factor of about 0.5 to 2, so a range of roughly CHF 32'000 to CHF 126'000 should be communicated. The estimate assumes one cross platform codebase, an existing backend API, and standard sensor access without custom drivers.